SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4 **Total Marks:40**

Annexure A

ARTICLE REVIEW

Code of Conduct:

I **C. PRIYANKA (192511074)** *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **C. PRIYANKA**

Annexure A

ARTICLE REVIEW

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications.*

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

(a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.

(b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

(a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.

(b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

(a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.

(b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?

(c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

(a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.

(b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

ANSWER

Research Article Review

Article Title

A comparison of machine learning algorithms for diabetes prediction

Task 1 – Article Summary

(a) Citation Details

Authors: Jobeda Jamal Khanam and Simon Y. Foo

Title: A comparison of machine learning algorithms for diabetes prediction

Journal: ICT Express

Volume: 7, Issue 4

Pages: 432–439

Year: 2021

Publisher: Elsevier

DOI: 10.1016/j.icte.2021.02.004

Domain/Application Area: Healthcare and medical diagnosis.

ML Problem: The paper addresses the problem of predicting diabetes using patient medical information. It is a supervised binary classification problem in which the model predicts whether a patient is diabetic or non-diabetic. (ScienceDirect)

(b) Key Objective and Research Gap

The main objective of the research is to compare different machine learning algorithms for predicting diabetes at an early stage. Diabetes is a serious disease, and early prediction can help patients receive proper medical attention. The authors used the Pima Indian Diabetes dataset and tested several machine learning techniques, including Decision Tree, K-Nearest Neighbour, Random Forest, Naïve Bayes, AdaBoost, Logistic Regression and Support Vector Machine. They also investigated neural network models with different numbers of hidden layers and training epochs. (Korea International Cooperation)

The research gap identified by the authors was that different machine learning methods may provide different prediction performances, and it was necessary to compare them systematically for diabetes prediction. The authors attempted to fill this gap by evaluating multiple classification algorithms on the same dataset. Their results showed that Logistic Regression and SVM performed well, while a neural network with two hidden layers achieved an accuracy of approximately 88.6%. (Korea International Cooperation)

Task 2 – Methodology Analysis

(a) ML Techniques Used

The paper uses several supervised machine learning classification algorithms:

1. **Decision Tree (DT)**
2. **K-Nearest Neighbour (KNN)**
3. **Random Forest (RF)**
4. **Naïve Bayes (NB)**
5. **AdaBoost (AB)**
6. **Logistic Regression (LR)**
7. **Support Vector Machine (SVM)**

8. **Neural Network (NN)**

The main CO4 statistical learning techniques are **Logistic Regression, Naïve Bayes and SVM**.

Dataset Used

The researchers used the **Pima Indian Diabetes (PID) dataset**, obtained from the **UCI Machine Learning Repository**.

- **Number of patients:** 768
- **Attributes:** 9 columns including the target/output
- **Input medical features:** Pregnancies, Glucose, Blood Pressure, Skin Thickness, Insulin, BMI, Diabetes Pedigree Function and Age.
- **Output:** Diabetes outcome.
- **Problem type:** Binary classification.
- **Preprocessing:** The data was prepared using data-mining/preprocessing procedures before applying the machine learning algorithms. The study used the Weka environment for preprocessing and model evaluation. (PubMed Central (PMC))

Basic Working

Patient Data → Preprocessing → Feature Analysis → ML Algorithms → Prediction → Performance Comparison

(b) Mapping to CO4 Topics

CO4 Topic	Technique Used	Application
Statistical Learning	Logistic Regression	Diabetes classification
Statistical Learning	Naïve Bayes	Diabetes classification
Statistical Learning	SVM	Diabetes classification
Decision Trees	Decision Tree	Diabetes classification
Inductive Learning	ML classifiers learn patterns from training data	Prediction
Reinforcement Learning	Not used	Not applicable

Methodological Strength

A major strength is that **multiple machine learning algorithms were compared using the same dataset**. This makes it easier to understand which algorithms are more suitable for diabetes prediction. The study also investigated different neural-network configurations. (Korea International Cooperation)

Methodological Limitation

The main limitation is the **small and single-source dataset**. The Pima dataset contains only 768 records, so the results may not generalize to patients from different populations or hospitals. Also, accuracy alone is not sufficient for a medical diagnosis problem.

Task 3 – Results and Critical Evaluation

(a) Key Results

The paper evaluated seven traditional ML algorithms and neural networks. Reported accuracy values summarized in later literature based on the study are approximately:

Algorithm	Accuracy
Decision Tree	74.24%
Random Forest	77.14%
Naïve Bayes	78.28%
Logistic Regression	78.85%
KNN	79.42%
AdaBoost	79.42%
SVM	77.71%
Neural Network	**88.60%**

The study reported that **Logistic Regression and SVM performed well among the traditional methods**, while the neural network with two hidden layers achieved about **88.6% accuracy**. (PubMed Central (PMC))

(b) Critical Evaluation

The reported results are reasonable for the Pima dataset, but they should not be interpreted as proving that the models will achieve the same performance in hospitals.

Are the metrics realistic?

Yes. Accuracies around 75–89% are realistic for a relatively small diabetes dataset. However, the 88.6% neural-network result should be interpreted carefully because performance can change depending on the train-test split and preprocessing.

Possible Biases

1. **Dataset bias:** The Pima dataset represents a specific population and may not represent all patients.
2. **Small dataset:** Only 768 records were available.
3. **Class imbalance:** The number of diabetic and non-diabetic samples is not equal.
4. **Evaluation limitation:** A single dataset may not demonstrate real-world generalization.
5. **Metric limitation:** Accuracy alone does not show false negatives, which are especially important in medical diagnosis.

Additional Experiments

The research could be strengthened by:

- Using larger hospital datasets.
- Testing the models on external datasets.
- Performing repeated stratified cross-validation.
- Reporting precision, recall, F1-score and AUC-ROC.
- Performing statistical significance tests.
- Using explainable-AI methods such as SHAP.

- Testing the models on patients from different geographical populations.

(c) Real-World Applicability

The proposed approach can be used in:

1. **Hospital decision-support systems**
2. **Diabetes screening applications**
3. **Primary healthcare centres**
4. **Mobile health applications**
5. **Remote patient monitoring**
6. **Electronic health record systems**

Challenges in Industry

Data availability: Large, high-quality and properly labelled medical datasets are difficult to obtain.

Privacy: Patient information must be protected.

Generalization: A model trained on one population may not work equally well for another population.

Scalability: The model must process large numbers of patient records efficiently.

Ethical issues: Incorrect predictions can affect medical decisions, so ML should support doctors rather than completely replace them.

Explainability: Healthcare professionals need to understand why a model predicted that a patient is at high risk.

<u>Task 4 – Reflection and Learning Outcome</u>

(a) Three Key Insights

Insight 1 – Logistic Regression

I learned that **Logistic Regression is useful for binary classification problems** such as predicting whether a patient has diabetes or not.

CO4 Connection: Statistical Learning Methods – Logistic Regression.

Insight 2 – SVM Classification

I learned that **Support Vector Machine separates classes by finding an effective decision boundary**. It can be useful when the relationship between input features and classes is complex.

CO4 Connection: Statistical Learning Methods – SVM.

Insight 3 – Model Comparison is Important

I learned that one ML algorithm cannot always be considered the best for every problem. Different algorithms can produce different results on the same dataset. Therefore, comparing models using suitable evaluation metrics is important.

CO4 Connection: Statistical Learning and Decision Tree-based classification.

(b) Proposed Extension

If I extend this research, I would use a **larger and more diverse diabetes dataset from multiple hospitals** and compare Logistic Regression, SVM, Random Forest and XGBoost.

I would also use **SHAP-based explainable AI** to identify which patient features contribute most to the prediction.

Proposed Experiment

Pima Dataset + Hospital Dataset → Data Cleaning → Feature Selection → LR/SVM/RF/XGBoost → Cross-Validation → Accuracy + Precision + Recall + F1 + AUC → Explainable AI

Expected Impact

This experiment would improve the reliability and generalization of the model. Using multiple datasets would reduce dataset bias, while explainable AI would make the predictions easier for doctors to understand. This could make the system more suitable for real-world diabetes screening.

Conclusion

The article demonstrates how different machine learning techniques can be applied to diabetes prediction. Logistic Regression, SVM, Decision Tree, Naïve Bayes and other classifiers were compared using the Pima Indian Diabetes dataset. The study showed that traditional statistical learning methods can provide useful prediction performance, while the tested neural network achieved about 88.6% accuracy. (ScienceDirect)

The article helped me understand the practical application of **Statistical Learning, Decision Trees and supervised classification**, which are important CO4 concepts. However, larger and more diverse datasets, additional evaluation metrics and external validation are required before deploying such a system in real healthcare environments.